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Four-View Visual–LiDAR Loop Closure for Drift Correction in Multi-Storey Indoor Mapping

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Abstract

Accumulated drift in visual–LiDAR mapping can produce duplicated corridor surfaces and distorted floor spacing in multi-storey buildings. Correcting these errors requires recognizing revisits despite changes in viewing direction and similar layouts on different floors. We present a loop-closure backend that combines images from four fisheye cameras with light detection and ranging (LiDAR) data. It searches for revisits by testing cyclic camera assignments, then estimates relative poses from cross-camera matches with LiDAR depth. Bidirectional geometric checks and consistency across successive revisit pairs screen these measurements before trajectory optimization. A shared height variable for each floor constrains the vertical structure of the map. On a single-storey recording, the corrected map reduced the median ground-height gap between two passes through a 15.5 m corridor from 21.00 to 0.39 cm. On a four-storey recording, the method accepted 26 loops across all four floors, all judged correct by visual inspection. The forward camera pair alone retained no loops. With the same 26 loops, adding floor constraints reduced the fourth-floor P95–P5 ground-height spread from 76.33 to 2.98 cm.

Keywords: indoor mapping; multi-camera loop closure; omnidirectional place recognition; visual–LiDAR fusion; multi-storey mapping; floor-aware pose graph.

1. Introduction

Indoor mapping with a multi-sensor platform involves traversing corridors, rooms, and stairwells and revisiting parts of the same building. Reconstructing the building consistently requires the system to recognize surfaces seen on earlier passes and recover their positions relative to the surrounding structure. This becomes difficult when a change in traversal direction changes which camera sees each surface and similar layouts recur at different heights. Drift shifts the estimated location of a revisit, while an incorrect association can pull distinct corridors or floors together during map correction. The central challenge is to identify revisits and measure their geometry reliably enough to maintain the building's overall structure.

Visual and inertial observations provide motion estimates and connections to previously mapped areas, as demonstrated by ORB-SLAM3 [1]. LiDAR–inertial methods organize geometric and motion measurements through smoothing or filtering, as in LIO-SAM and FAST-LIO2 [2, 3]. FAST-LIVO2 uses LiDAR geometry to support direct image alignment, bringing appearance and range information into the same estimation process [4]. Multi-camera systems extend this information to more viewing directions: Omni-LIVO uses cross-view photometric observations, and Multi-LVI-SAM organizes fisheye observations within a panoramic model [5, 6]. These developments provide the trajectories, depth, and surrounding views needed to connect observations throughout an indoor survey.

Multi-sensor fusion connects motion estimation with consistent reconstruction across complex environments. Hao et al. fuse LiDAR, inertial measurements, and wheel odometry in a filtering frontend and use spherical-target constraints in a pose-graph backend for underground mapping [7]. For multi-floor indoor localization, Cheng et al. combine WiFi fingerprints, stacked-autoencoder features, and multi-label classification to estimate building, floor, and position [8]. Long corridors remain a demanding part of this task because the observed structure can constrain some motion directions weakly [9]. Direct visual–LiDAR fusion adds image-alignment constraints to the geometric measurements, providing another source of motion information [10]. Reobserving an earlier location also provides a connection across time. Systems such as RTAB-Map use place retrieval and geometric verification to establish these connections for graph optimization [11]. Factor graphs provide a common formulation for the resulting measurements [12]. The usefulness of a revisit therefore depends on both its place association and the relative pose assigned to it.

Opposite-direction revisits change the camera correspondence needed for place association. A wall seen by a forward-facing camera on the outward pass may appear in a rear camera on the return pass. Sequence-based retrieval, including SeqNet, uses consecutive observations to strengthen place recognition [13]. Local features and learned matching then provide image correspondences that can be tested geometrically [14, 15]. With a surrounding camera rig, these steps also need to account for the transfer of observations between physical cameras. Carrying the selected camera correspondence from retrieval into pose estimation links the evidence used to recognize a revisit with the evidence used to measure it.

Repeated layouts on different floors create another source of ambiguity. LiDAR place descriptors use spatial structure and local geometric arrangements to retrieve candidate locations [16, 17]. Their representations also address changes in heading and position through structural matching and invariant geometric relationships [18, 19]. This makes LiDAR retrieval valuable when the image appearance changes substantially. In a building with similar corridors on several levels, however, local structural similarity must be interpreted together with the route and floor identity. Successive revisit measurements provide an opportunity to check whether a set of locally plausible matches agrees on how the trajectory should be corrected.

Robust optimization and ground constraints offer complementary ways to use this information. GNC adjusts the influence of measurements according to their residuals in a robust estimation problem [20]. Ground-SLAM associates ground-plane observations across keyframes and incorporates them into a pose graph for multi-floor mapping [21]. These approaches motivate two connected checks in our setting: revisit

measurements should agree along the route, and the corrected poses should remain consistent with the observed floor surfaces. A loop-closure backend can connect these checks by retaining camera correspondence during verification and by linking ground observations throughout each floor.

We develop this backend using the trajectory, four-camera images, and LiDAR submaps supplied by Omni-LIVO [5]. Cyclic camera assignments match the surrounding views across visits, and LiDAR depth turns the pooled cross-camera correspondences into relative-pose measurements. Bidirectional LiDAR checks examine the local alignment. A route-sequence test then compares the corrections implied by neighbouring pairs, and a floor-aware graph combines the selected loops with ground observations. Each floor has a shared height variable, allowing observations made at different positions and times to contribute to its vertical estimate.

The main contributions are:

A joint four-view loop-closure method that carries cyclic camera correspondence from place retrieval into geometric pose verification. Similarity normalization for each camera pair and floor compensates for camera-pair score bias, while joint scoring over cyclic assignments combines the four views to associate places across changes in viewing direction, including reverse revisits. The selected assignment guides cross-camera feature matching, and LiDAR-associated correspondences from multiple cameras jointly constrain a common body-frame relative pose, linking surrounding visual observations to metric loop-closure estimation.

A route-sequence verification method that evaluates geometrically verified revisit pairs through their implied trajectory corrections in a common world frame. Direction-aware sequence construction accommodates forward and reverse traversal, and complete-link consistency requires mutual agreement across spatially extended, same-floor observations. Spatial anchor selection prioritizes pairs with more pose inliers and removes temporal and spatial redundancy, providing route-consistent loop constraints for graph optimization.

A floor-aware robust pose graph that jointly estimates trajectory poses and shared floor heights to constrain within-floor deformation and inter-floor height drift. Ground-plane factors link pose orientation and elevation across observations of the same floor, while measurements between consecutive stable segments connect unequal floor separations. These structural factors are optimized with odometry and loop constraints, using geometry-dependent loop uncertainty and graduated non-convexity to regulate the influence of unreliable loop measurements.

The evaluation uses single-storey and four-storey recordings, controlled ablations, external loop-closure configurations, and a public multi-camera retrieval sequence.

2. Related Work

2.1. LiDAR–visual–inertial odometry with multiple cameras

Multisensor mapping systems differ in how they exchange information between estimation stages. LIO-SAM organizes inertial, LiDAR, and historical constraints within a factor graph [2], while FAST-LIO2 directly registers LiDAR points within an iterated filtering framework [3]. LVI-SAM illustrates how the modalities can assist one another: LiDAR supplies depth for visual estimation, visual estimates initialize scan matching, and visually detected loops are refined by the LiDAR subsystem [22]. FAST-LIVO and FAST-LIVO2 use direct image alignment with LiDAR geometry [4, 10]. R3LIVE uses its LiDAR–inertial subsystem to build map geometry and its visual–inertial subsystem to recover colour [23]. These designs link motion estimation with reconstruction by making measured depth available to visual processing and using image information to enrich geometric estimates.

Multiple cameras add observations from directions that a single camera cannot cover simultaneously. Offermann combines learned image preprocessing with geometric matching for indoor place recognition using panoramic images [24]. Camera calibration provides the geometric representation needed to combine observations across views [25]. Omni-LIVO extends direct photometric estimation across camera views through temporal migration and multi-view filtering updates [5]. Multi-LVI-SAM takes a panoramic representation approach: it combines fisheye features, compensates for camera-centre offsets during triangulation, retrieves loop candidates, and passes them to LiDAR verification and graph optimization [6].

Our backend builds on these capabilities by retaining the physical camera identities during revisit retrieval. The selected cyclic assignment specifies the camera pairs used in the relative-pose estimate. Verification can therefore use the view pairs that produced the retrieval score, and sequence checking compares the resulting metric corrections along the route. Omni-LIVO supplies the input observations and initial trajectory used in our experiments.

2.2. Place recognition under viewpoint change and perceptual aliasing

Visual place descriptors compress an image into a representation that can be compared with earlier observations. One design choice is how to combine local image information. NetVLAD aggregates local descriptors through trainable assignments to cluster centres [26]. MixVPR models spatial relationships within backbone feature maps through feature mixing [27]. The two approaches learn different aggregation operations for the same purpose: producing an image descriptor that distinguishes places.

The source of the local features also affects how retrieval can be adapted to new environments. DINOv2 supplies visual representations learned from a large image collection through self-supervision [28]. AnyLoc combines pretrained features with unsupervised aggregation, allowing a retrieval descriptor to be formed without labelled place pairs [29]. SALAD uses optimal transport to learn feature assignments and includes an additional cluster for less informative features [30]. It combines pretrained representations with aggregation learned for place recognition. GeM offers a simpler pooling operation between average and

maximum aggregation [31]. Together, these approaches provide choices for constructing each image descriptor. In a multi-camera system, the pairing and combination of those descriptors then determine how simultaneous views contribute to a candidate score.

The spatial arrangement of the sensors and the order of observations provide information beyond an individual image descriptor. SeqNet learns short-sequence representations and uses them in hierarchical place retrieval [13]. In our method, cyclic camera assignments preserve the layout of the rig when image scores are combined. The selected assignment also identifies the views used for pose verification. The later sequence test operates on the trajectory corrections inferred from verified relative poses. Temporal information is therefore used to assess the agreement of the resulting measurements along the route.

LiDAR descriptors offer different ways to preserve scene structure across viewpoint changes. Scan Context records spatial structure in radial and angular bins and searches circular shifts to accommodate heading changes [16]. Scan Context++ extends this treatment to lateral variation [18]. STD uses triangle geometry to establish correspondences under rigid motion [19], and BTC combines triangle structure with local binary descriptions to improve distinctiveness [17]. These triangle-based methods can supply correspondences for pose verification as well as retrieval scores. BTC has also been evaluated under reverse traversal and large viewpoint changes [17]. OverlapNet predicts scan overlap and relative yaw, supplying information useful for candidate selection and subsequent alignment [32]. These outputs help bridge the steps of recognizing a place and aligning its observations. In repeated indoor layouts, our verification additionally relates each candidate to its floor and to neighbouring measurements along the traversal route.

2.3. Loop verification and robust pose-graph optimization

Geometric verification connects a retrieved place to a usable pose measurement. SuperPoint provides local keypoints and descriptors [14]. SuperGlue uses context from the two feature sets to establish correspondences [33], and LightGlue adapts the matching computation to the difficulty of the image pair [15]. RANSAC evaluates whether subsets of these correspondences support a common model [34]. Five-point estimation recovers relative geometry between two calibrated views [35]; AP3P estimates a camera pose from known 3D points and image bearings [36]. LiDAR depth makes the latter formulation available in our pipeline. We generate hypotheses for individual cameras, transform them into a shared body frame, and score them with correspondences from multiple cameras.

Point-to-plane registration refines the alignment using the local surfaces represented by the LiDAR submaps [37]. Its uncertainty affects how strongly the resulting loop should influence the trajectory. Censi analyses ICP covariance with explicit attention to measurement noise and correlation [38]. This work motivates treating a loop as an uncertain geometric measurement. Our implementation uses a Hessian-based approximation with additional regularization; its construction is described in Section 3.6.

The verified measurements are then combined with odometry in a pose graph [12]. RTAB-Map provides an established example of connecting appearance retrieval, geometric verification, and graph optimization [11]. GNC addresses inconsistent measurements by updating their weights within a robust objective [20]. These weights depend on residuals in the current graph. Our route-sequence test precedes this step and

checks whether neighbouring loops imply compatible corrections to the initial trajectory. The two stages assess consistency at different scales: along a revisited route segment and across the optimized graph.

2.4. Ground constraints and multi-floor mapping

Ground observations provide a structural reference that can be shared across different parts of a trajectory. Ground-SLAM represents ground surfaces as plane landmarks, uses a closest-point parameterization, and associates observations of the planes across keyframes [21]. Incorporating these associations into a pose graph lets ground geometry constrain poses beyond an individual scan. This is directly relevant to buildings, where observations collected at different corridor locations can belong to the same physical floor.

We use one scalar height variable for each identified floor. Each ground-plane observation aligns the ground normal with the vertical direction. It also links the ground height implied by its associated pose to the shared floor height. Consecutive stable segments supply measured connections between floor levels, allowing adjacent floors to have different separations. This parameterization groups observations by their role in the building's vertical structure and combines them with odometry and loop factors in a common graph [12]. It also allows ground observations from singly traversed segments to contribute to the correction.

3. System and Method

3.1. Multi-sensor platform

The custom unmanned aerial vehicle (UAV) has a 0.60 m wheelbase, a take-off mass of 4.4 kg, and a Radiolink PIX6 flight controller (Figure 1a). The flights use position-assisted control at approximately 1.5 m/s. The sensing platform comprises a Livox MID-360 LiDAR with a built-in inertial measurement unit (IMU), four SC132GS global-shutter cameras with 206° fisheye lenses, an STM32 synchronization board, and an onboard computer. The board triggers all cameras at 20 Hz and supplies the time reference for sensor alignment.

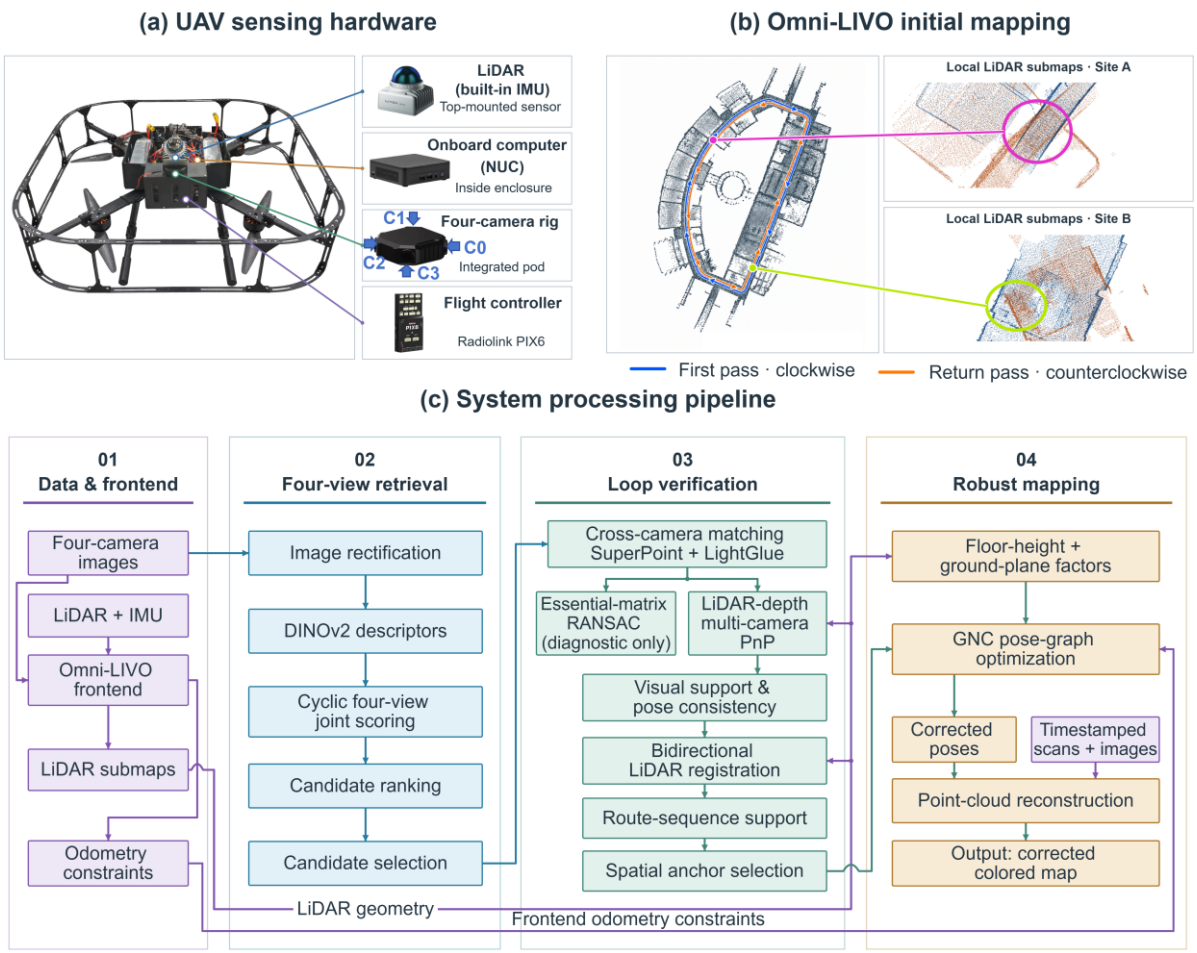


Figure 1. Multi-sensor platform and processing pipeline. (a) Sensing hardware; the four-camera assembly is beneath the LiDAR. (b) Initial Omni-LIVO map and local LiDAR submaps at two revisit sites. (c) Four-view retrieval, multi-camera pose and LiDAR verification, route-sequence selection, and floor-aware GNC optimization.

The cameras are indexed by their physical positions. Cameras c0 and c1 face front-left and front-right, respectively, and form the forward pair. Cameras c2 and c3 face rear-right and rear-left. These indices remain fixed throughout retrieval and verification.

Each camera is calibrated at a resolution of 1088 × 1280 pixels. The calibration includes focal lengths, principal points, the unified-camera-model parameter ξ, and four distortion coefficients [25]. FAST-Calib provides the camera–LiDAR extrinsics [39]. The camera–LiDAR and LiDAR–IMU transforms express observations in a common body frame.

3.2. Problem formulation and processing overview

The initial trajectory, four-camera images, and LiDAR submaps are supplied by Omni-LIVO [5]. Each node i contains an initial body pose $T_i^0 \in SE(3)$, a timestamp t_i, four images I_i^c with $c \in \{0,1,2,3\}$, and a local LiDAR submap P_i. We write the corrected body-to-world pose as $T_i = [R_i, p_i]$. The backend selects a set of loop constraints L and estimates the corrected poses together with a height h_f for each identified floor. The frontend supplies the consecutive motion measurements and the calibration used to associate image observations with LiDAR geometry.

Figure 1(c) shows the processing stages. Retrieval first proposes candidate pairs. A pair becomes verified after passing the visual, odometry, and LiDAR checks. It gains sequence support when neighbouring pairs imply compatible trajectory corrections. Redundancy removal then selects spatial anchors from the supported pairs. The anchors retained by the final graph optimization are the accepted loops.

3.3. Four-view retrieval under cyclic camera correspondence

Each fisheye image is rectified into three 512 × 512 pinhole views. The views have a 90° field of view and yaw offsets of −50°, 0°, and +50°. The resulting 12 views cover the surroundings of the camera rig. Masks exclude invalid image regions and visible parts of the airframe.

We encode each view with the DINOv2 vision transformer ViT-B/14 [28]. Valid patch tokens are L2-normalized and clipped to a minimum of 10^{-6}. Generalized-mean (GeM) pooling with exponent 3 then combines them into a single vector [31]. We separately L2-normalize this vector and the class token, concatenate them, and normalize the result. For nodes i and j and physical cameras a and b, the camera-pair similarity is defined by Equation (1).

$$C_{ij}(a,b) = \max_{v,w} d_{i,a,v}^{\top} d_{j,b,w}, \tag{1}$$

The indices v and w denote the three rectified views from each camera. The maximum selects the strongest match between these views, allowing the comparison to use the part of each camera image that best captures the shared surface.

Similarity scores can differ between camera pairs. We standardize each score using the mean and standard deviation for that camera pair on floor f, as shown in Equation (2).

$$\tilde{C}_{ij}(a,b) = \frac{C_{ij}(a,b) - \mu_{f,a,b}}{\max(\sigma_{f,a,b}, 10^{-6})}. \tag{2}$$

The statistics use same-floor pairs that satisfy the temporal exclusion rule. We use floor-specific statistics when at least 32 eligible pairs are available. Otherwise, we use statistics for the same camera pair over the full recording. Floor labels are obtained from ground observations and the initial trajectory before retrieval, as described in Section 3.6.

The joint score evaluates four cyclic assignments of the physical cameras (Equation (3)).

$$S(i,j) = \max_{s \in \{0,1,2,3\}} \frac{1}{4} \sum_{c=0}^{3} \tilde{C}_{ij}\,(c, (c+s) \bmod 4). \tag{3}$$

Each shift pairs the cameras at one visit with a corresponding rotation of the camera order at the other visit. We retain the highest score and pass its camera assignment to pose verification. During a reverse revisit, a forward camera can therefore match a rear camera from the earlier visit. The pose solver subsequently estimates the continuous rotation and translation between the two body frames.

Pairs separated by less than 35 s are excluded. Each query retains up to 12 candidates from the same floor and three with uncertain floor membership. At most two candidates are selected from any 12 s interval of the historical recording. Pairs known to lie on different floors are excluded before verification. Candidates

with uncertain membership can proceed to verification; final loop acceptance requires both endpoints to have the same floor label. Figure 2(a–c) illustrates the retrieval process and score normalization.

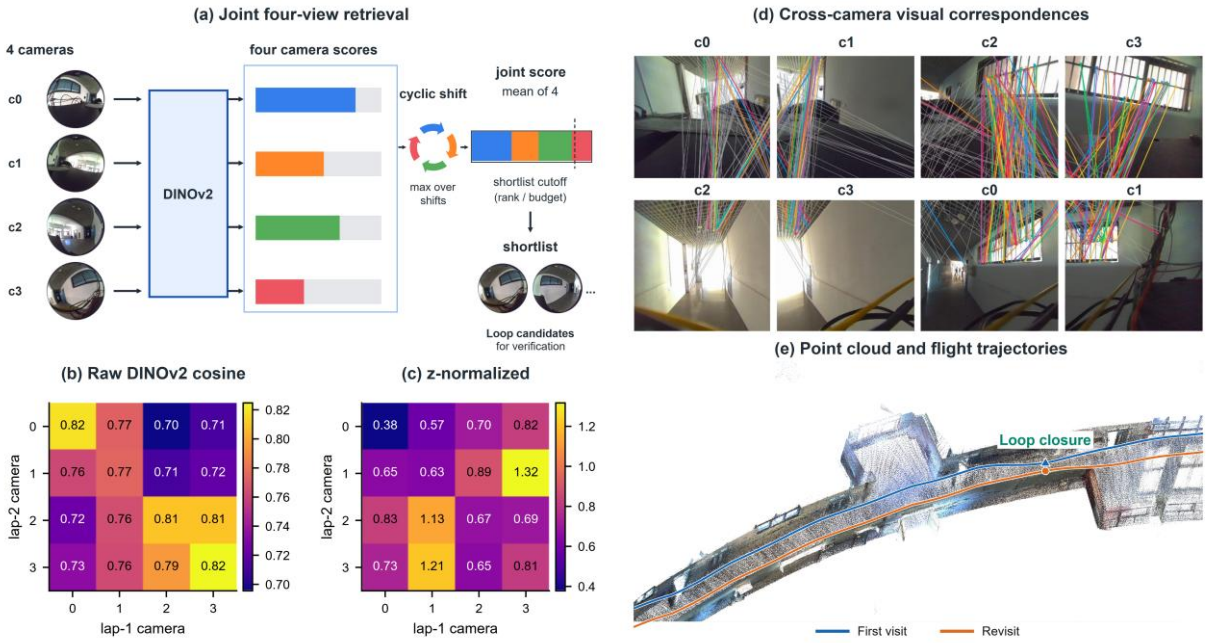


Figure 2. Four-view retrieval, camera-pair normalization, and reverse-revisit correspondences. (a) Descriptor extraction, cyclic assignment, joint scoring, and candidate selection. (b) Raw DINOv2 cosine similarities and (c) normalized camera-pair scores, averaged over 47 strict-positive pairs in a single-storey diagnostic with 56 query and 56 database nodes. The heatmaps show score distributions; the query-level retrieval evaluation in Section 4.2 uses 44 certain-positive queries. (d) First-visit and revisit images from the four-camera rig, arranged by cyclic shift $s = 2$: c0–c2, c1–c3, c2–c0, and c3–c1. Colored lines indicate essential-matrix RANSAC inliers; grey lines indicate matches rejected by that diagnostic. (e) Single-storey coloured point cloud with first-visit and revisit trajectories and the loop location.

3.4. Cross-camera pose verification with LiDAR depth

For each selected physical-camera pair, we use SuperPoint and LightGlue to match the two highest-scoring pairs of rectified views [14, 15]. The matched observations are converted to calibrated bearing vectors, merged across rectified views, and deduplicated. Essential-matrix RANSAC provides the match labels shown in Figure 2(d) [34, 35]. The subsequent perspective-n-point (PnP) stage uses all deduplicated correspondences with valid LiDAR depth, including matches rejected by that image-based diagnostic.

LiDAR depth associates each historical bearing with a 3D point in the historical body frame. The corresponding query observation is a bearing in its camera frame. Algebraic perspective-three-point (AP3P) within RANSAC generates pose hypotheses for each camera [34, 36]. We transform these hypotheses into the common body frame and score them using correspondences from all participating cameras. A match is an inlier when its angular error is at most 0.8°. The selected pose is refined using Huber-weighted angular residuals.

A candidate requires at least 15 PnP inliers, with at least five inliers in each of two or more cameras. The estimated relative pose must also agree with the frontend odometry within 5 m and 20°.

We then refine the pose by point-to-plane iterative closest point (ICP) registration, initialized from the PnP estimate [37]. The refined transformation is evaluated in both directions between the two submaps. Each direction requires coverage of at least 0.25 and a point-to-plane root mean square error (RMSE) of at most 0.18 m. Evaluating both directions checks that the alignment covers a sufficient part of each submap. The two evaluations use the same refined transformation.

The registration update is limited to 0.5 m and 5° relative to the PnP estimate. We also check that the transformation is finite, its rotation is orthogonal within a tolerance of 10^{-5}, its determinant is at least 0.99, and its final row has the homogeneous-transform form. A candidate that passes these checks becomes a verified pair for sequence analysis.

Coverage is computed after 0.2 m voxel sampling. A source point contributes when it has a target neighbour within 0.3 m and the target neighbourhood has planarity above 0.3. Reverse coverage uses the same rule with the source and target exchanged. ICP runs for up to 30 iterations, reduces the correspondence distance from 1.0 to 0.3 m, and weights each point-to-plane residual r by $1/[1 + (r/0.1)^2]$. Supplementary Table S3 summarizes these settings.

3.5. Route-sequence support and spatial anchors

The sequence test groups candidate pairs from successive observations of a revisited area. We test whether these pairs agree on the correction to the initial trajectory. Equation (4) expresses the correction implied by pair (i, j) in the world frame.

$$A_{ij} = T_j^0 \hat{T}_{j \leftarrow i} (T_i^0)^{-1}, \tag{4}$$

Here, $\hat{T}_{j \leftarrow i}$ maps query-body coordinates into the historical body frame. Expressing all pairwise measurements as world-frame corrections allows them to be compared along the route.

The sequence search considers traversal in both directions. Query timestamps always increase. Historical timestamps increase for a forward traversal and decrease for a reverse traversal. Consecutive query observations can be separated by at most τ seconds. The historical increment, measured in the corresponding traversal direction, can be at most 1.5τ. We use $\tau = 24$ s for both recordings, giving limits of 24 s on the query side and 36 s on the historical side.

A supported sequence contains at least three pairs from the same identified floor. On both sides of the sequence, the Euclidean distance between the first and last positions must be at least 6 m. We also require every pair of members to agree within 0.8 m and 8° in their correction transforms. This all-pairs requirement is the complete-link consistency test.

We select spatial anchors from the supported pairs in descending order of PnP inlier count. A candidate is redundant if both of its endpoint timestamps are within 12 s of those of an already selected pair and both endpoint positions are within 3 m. The remaining pairs become the anchors used for graph optimization.

3.6. Floor-aware robust pose graph

Ground observations provide shared structural constraints for multi-floor mapping [21]. We fit local ground planes using RANSAC [34] and identify stable floor segments from these fits and the initial trajectory before retrieval. Within a ±9 s window, the ground-height range must be at most 0.60 m and the fitted rate of height change at most 0.07 m/s. A segment requires at least five samples, a duration of 12 s, and 6 m of horizontal travel. Each ground-plane fit requires at least 150 inliers, an RMSE of at most 0.06 m, and a tilt of at most 12°.

Segments whose centre heights differ by no more than 0.9 m are assigned to the same floor. A separation of at least 1.8 m identifies another floor. Nodes with ambiguous membership remain unlabelled.

We use a factor graph to combine pose and ground observations [12]. The graph estimates the corrected body poses and one shared height for each identified floor. Let f(i) be the floor assigned to node i. The fitted ground plane in the body frame has a unit normal and is written as Equation (5).

$$(n_i^b)^\top x_b + d_i^b = 0, \tag{5}$$

After applying the corrected rotation, the world-frame normal is $n_i = R_i n_i^b$. Let z_i be the vertical component of the corrected position p_i. Equation (6) defines the ground residual.

$$r_i^g = \begin{bmatrix} n_{i,x} \\ n_{i,y} \\ z_i - d_i^b / n_{i,z} - h_{f(i)} \end{bmatrix}. \tag{6}$$

The first two components align the ground-plane normal with the vertical direction. The third compares the plane's estimated height directly beneath the sensor with the shared height of its floor. We apply a Huber loss to the norm of the whitened ground residual [12]. Thus, each ground observation constrains its own pose while also linking that pose to the other observations assigned to the same floor.

Consecutive stable segments provide measurements between floor levels. Their initial ground-centre height difference defines Δ_{kl}, with residual $h_l - h_k - \Delta_{kl}$. The measurement standard deviation is the larger of 0.3 m and the mean of the two segments' P90–P10 height spreads. Each floor height also has a Gaussian prior centred at its initially identified level, with a standard deviation of 0.5 m. The measured differences allow adjacent floors to have unequal separations.

Odometry factors preserve the consecutive relative transformations from the frontend. Loop factors connect revisited poses using the verified relative-pose measurements. Equation (7) gives the loop residual.

$$r_{ij}^\ell = \mathrm{Log}\left(\hat{T}_{j\leftarrow i}^{-1} T_j^{-1} T_i\right). \tag{7}$$

We approximate the loop covariance from the inverse point-to-plane ICP Hessian, following the local least-squares covariance analysis in [40]. We use a residual scale of max(RMSE, 0.02 m) and add diagonal variances of (1°)² for rotation and (0.10 m)² for translation. If registration covariance is unavailable, the initial diagonal standard deviations are 2° and 0.2 m. After transforming the covariance into the graph residual frame, we add further diagonal variances of (2°)² and (0.10 m)².

We use the Hessian eigenstructure to identify weakly constrained translation directions [9]. A normalized minimum eigenvalue below 0.03 or a condition number above 25 indicates degeneracy. In that case, the variance along the weakest direction is set to at least (0.5 m)². The information matrix is the inverse of the regularized covariance. For a group containing n anchors from the same floor and sequence, we multiply each information matrix by min(1, 3/n). This reduces the weight of each anchor in groups with more than three members.

The SE(3) logarithm in Equation (7) uses rotation-first coordinates [ω; v]. The first pose is fixed by a gauge prior. Equation (8) combines odometry, loop, ground, inter-floor, and floor-height prior terms into the optimization objective.

$$\begin{aligned}\min_{\{T_i\},\{h_f\}} \quad & \sum_{i=1}^{N-1} \| r_i^o \|_{\Omega_o}^2 + \sum_{(i,j)\in\mathcal{L}} w_{ij} \| r_{ij}^{\ell} \|_{\Omega_{ij}}^2 \\ & + \sum_i \rho_H \left(\| r_i^g \|_{\Omega_g} \right) + \sum_{(k,l)} \frac{(h_l - h_k - \Delta_{kl})^2}{\sigma_{kl}^2} + \sum_f \frac{(h_f - h_f^0)^2}{\sigma_h^2}.\end{aligned} \tag{8}$$

Here, r_i^o is the consecutive-pose odometry residual and $\sigma_h = 0.5$ m. GNC updates the loop weights from their squared whitened residuals q_{ij} [20]. We use a cutoff $c^2 = 16.81$, corresponding to the 99th percentile of a chi-squared distribution with six degrees of freedom. Equation (9) gives the truncated-least-squares weight update.

$$w_{ij} = \begin{cases} 1, & q_{ij} \le \frac{\mu}{\mu+1}\bar{c}^2, \\ \sqrt{\frac{\bar{c}^2\mu(\mu+1)}{q_{ij}}} - \mu, & \frac{\mu}{\mu+1}\bar{c}^2 < q_{ij} < \frac{\mu+1}{\mu}\bar{c}^2, \\ 0, & q_{ij} \ge \frac{\mu+1}{\mu}\bar{c}^2. \end{cases} \tag{9}$$

We initialize μ as $\max\{c^2 / \max[2\max(q) - c^2, 10^{-6}], 10^{-4}\}$ and multiply it by 1.4 at each update, for up to 60 updates. Optimization stops when every weight is within 10^{-3} of either 0 or 1 and $\mu > 50$. Each weighted graph is solved with Levenberg–Marquardt [12]. We allow up to 100 iterations. The first-pose prior has a standard deviation of 10^{-4}. Loops with a final weight of at most 0.5 are removed, and the graph is solved again with unit weights for the retained loops. Supplementary Table S3 gives the odometry and ground noise settings.

3.7. Map reconstruction and implementation

The corrected poses transform the timestamped observations into the final map. For the single-storey comparison, both reconstructions use the same full-resolution coloured points in the same order. Changes in repeated-surface alignment can therefore be traced to the pose correction. We also export the corrected four-storey coloured map to inspect the reconstructed building structure.

The backend runs offline on an ASUS ROG Strix G713PV computer with an AMD Ryzen 9 7845HX CPU, an NVIDIA GeForce RTX 4060 Laptop GPU, and 16 GB RAM. Both indoor experiments use PyTorch 2.14.1, OpenCV 5.0.0.93, and Georgia Tech Smoothing and Mapping (GTSAM) 4.3.0. Supplementary Table S3 lists the processing thresholds and solver settings.

The Supplementary Methods, “Reproducibility” provides the experiment scripts, configuration details, and instructions for reproducing the results.

During manuscript preparation, the authors used Codex (OpenAI, desktop version 26.1023.6621.0) to assist with manuscript review, writing, and language refinement. The authors checked and revised the output and accept full responsibility for the final publication.

4. Experimental Design

4.1. Recordings and data

We evaluate the backend on two indoor recordings acquired using the same UAV-mounted sensing platform. The single-storey sequence retraces a route in the opposite direction, allowing us to measure alignment between the outward and return observations. It contains 134 nodes over 401 s along an approximately 485 m route. Retrieval produces 1338 candidate pairs, and the reconstructed map contains 153,108,192 coloured points.

The four-storey sequence connects corridors through stairwells and tests loop detection across floors together with the vertical structure of the map. We label the physical floors F1–F4 from lowest to highest. The route visits corridor segments on F3, F2, F1, F4, and F3, with stairway transitions between them. The recording lasts approximately 744 s and contains 243 nodes, 1100 candidate pairs, and 328,306,517 reconstructed coloured points. Its path length, computed from the frontend trajectory, is approximately 728.11 m. Individual LiDAR submaps contain 10,574–92,071 points.

We also evaluate retrieval on Quad-Easy from the multi-camera extension of the Newer College Dataset (NCD) [41, 42]. The sequence was collected outdoors with a handheld rig and lasts approximately 198 s. Our evaluation uses 66 nodes at approximately 3 s intervals and a common set of 31 retrieval queries. Ground-truth poses supply the input geometry for this experiment.

4.2. Evaluation metrics

We measure alignment between two passes by comparing surface positions in their reconstructed point clouds. The single-storey corridor evaluation uses outgoing nodes 14–20 and return nodes 115–120 over a 15.5 m interval. Ground heights are estimated in shared 0.5 m cells. The signed gap in each cell is the return-pass height minus the first-pass height. We report the median absolute gap over 206 cells present in both passes and both reconstructions.

For the longitudinal profile in Figure 3(c), the observations are grouped into 31 stations along the corridor. Upper-surface and wall-normal gaps are measured at 62 common locations. A separate 8 × 8 m region around the start and finish provides a broader ground comparison. The Supplementary Methods, “Single-storey region-wide alignment” gives the extraction rules for these measurements.

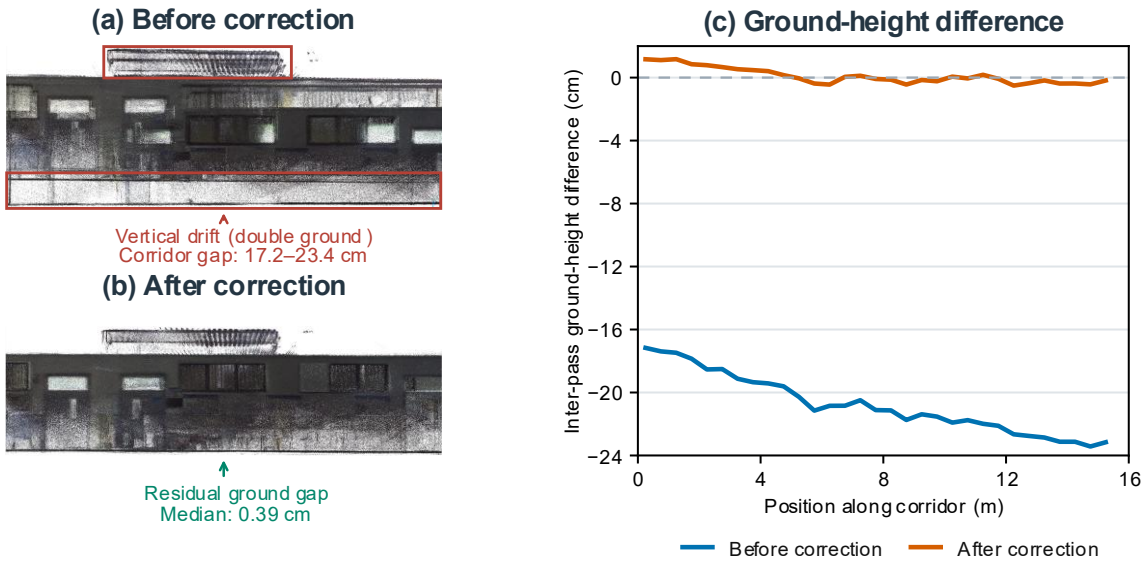


Figure 3. Two-pass alignment in the single-storey corridor beside the start/finish classroom. Side views (a) before and (b) after correction. (c) Signed ground-height difference, revisit minus first visit, at 31 stations spaced 0.5 m along the 15.5 m interval; zero marks coincident ground. Over 206 common cells, median absolute gaps are 21.00 and 0.39 cm. The 17.2–23.4 cm annotation in (a) indicates the absolute ground-height gap range before correction in (c).

One author reviewed accepted loops using synchronized four-camera images from both endpoints and their positions in the corrected map. A loop was labelled correct when both endpoints observed the same place on the same floor. This review assesses the place association of each accepted loop.

Retrieval recall at rank k, denoted R@k, is the fraction of eligible queries with a positive match among the top k candidates. We report R@1 and R@5. The single-storey benchmark contains 44 queries with confirmed positive matches and permits a ±1-node tolerance. The Supplementary Note, “Public-dataset retrieval” defines the public-dataset protocol, including its exact-node and tolerance-based rules.

For the four-storey map, adjacent-floor separation is the difference between the centres of transformed ground observations on the two floors. The P95–P5 ground-height spread within each floor measures its vertical consistency. Reference separations for F1–F2, F2–F3, and F3–F4 are 4.201, 4.255, and 4.169 m. These are the means of repeated measurements with a DLX-MD50 laser distance meter at the centres of the ground-observation regions. The instrument has a nominal accuracy of ±(2 mm + D/10000), where D is the measured distance expressed in consistent units.

For the robustness stress test, we report the largest displacement of any optimized node from its position in the input trajectory.

4.3. Compared methods

We compare loop-closure pipelines using the same four-storey recording processed by Omni-LIVO [5]. Each pipeline uses the retrieval, verification, and graph settings described below. The uncorrected frontend

provides the input poses and the reference reconstruction. Component ablations are described in Section 4.4.

For Scan Context [16], our comparison uses 20 rings, 60 sectors, an 80 m radius, three ring-key neighbours, and a cosine-distance cutoff of 0.20. It excludes the eight most recent nodes and applies a 15 m distance check before bidirectional ICP. Its graph combines consecutive frontend odometry with the accepted LiDAR loop measurements. We reconstruct the map with both Gaussian and Cauchy-robust loop-noise models.

For Real-Time Appearance-Based Mapping (RTAB-Map) [11], we use version 0.21.13 with the Omni-LIVO poses, rectified multi-camera images, and LiDAR-projected depth. We test three configurations: default PnP verification; depth-based 3D–3D registration with a 0.3 m inlier threshold and at least eight inliers; and the same 3D–3D registration with an additional 2.5 m distance restriction. For BTC [17], we use the same LiDAR submaps under four parameter combinations. Its candidates then pass through our same-floor, route-sequence, and odometry-consistency checks.

The forward-pair variant uses cameras c0 and c1 for both retrieval and verification, with the same candidate budget and thresholds as the four-view method. It scores both identity and swapped camera assignments and passes the selected assignment to verification. Independent and fixed-camera ranking variants change candidate scoring while retaining four-camera verification. The public-dataset comparison includes Scan Context [16], MixVPR [27], and DINOv2-based retrieval [28]. All methods use the common query protocol in the Supplementary Note, “Public-dataset retrieval”.

4.4. Ablation design

The camera-input and ranking experiments examine two uses of the four-camera observations. The first uses only the forward pair for retrieval and verification, while retaining the candidate budget and subsequent thresholds. The ranking experiments use independent or fixed-camera scoring to select candidates, with four-camera verification and the rest of the backend unchanged.

The LiDAR-verification ablation removes bidirectional geometric checking and registration refinement from both indoor experiments. LiDAR depth is still used for PnP, and the subsequent selection rules remain the same. The four-storey sequence-gap study varies the query-side interval from 12 to 48 s while keeping the candidate pairs and geometric observations fixed. A separate four-storey ablation disables GNC [20] within the main pipeline.

To examine the floor constraints, we keep the same 26 loops and the floor labels used during retrieval. We then remove the floor-height variables and their associated factors from the graph. The resulting changes in adjacent-floor separation and within-floor ground spread measure the contribution of those factors.

The single-storey sequence/GNC stress test uses a more permissive set of 55 pairwise measurements. Each pair has at least 10 PnP inliers and passes the ICP overlap and refinement checks, with no minimum camera-participation requirement. We test all four on/off combinations of sequence checking and GNC using the same measurements and covariances. Floor factors are disabled throughout this stress test.

5. Results

5.1. Single-storey sequence: alignment of the first and return passes

The method accepted 17 loops on the single-storey sequence, all judged correct by visual inspection. Of the 1338 candidate pairs, 64 passed pairwise verification and 60 gained sequence support. Anchor selection retained 17 pairs, all of which remained after graph optimization. On the 44-query retrieval benchmark, R@1 was 16/44 (36.4%) and R@5 was 34/44 (77.3%). The candidate budget of up to 15 per query allows pose verification to examine matches ranked below first place.

Figure 2(d,e) illustrates how cross-camera matching connects observations from opposite traversal directions. The corridor appears in different camera views during the two visits. The selected camera assignment brings those views into correspondence, and the point-cloud panel locates the revisit within the reconstructed corridor.

Correction brought the repeated ground observations into closer alignment. Across the 206 common cells in the 15.5 m corridor, the median absolute gap decreased from 21.00 to 0.39 cm, with 183 cells improving. Figure 3(a,b) shows the change in the duplicated surfaces, and Figure 3(c) plots the signed ground-height gap at 31 longitudinal stations. Both reconstructions contain the same 153,108,192 coloured points in the same order. The measured change reflects the combined effect of the loop and floor constraints.

The improvement also appears in the upper surfaces and walls. At 62 common locations, the median upper-surface gap decreased from 20.76 to 0.38 cm, and every location improved. The median wall-normal gap decreased from 3.73 to 1.76 cm, with 38 of 62 locations improving. This wall measure compares median lateral coordinates, which vary with occlusion and point density (the Supplementary Note, “Upper-surface and lateral-wall extraction”). In the broader 8 × 8 m start/finish region, the median ground gap decreased from 34.83 to 7.32 cm, with 97 of 108 cells improving (Supplementary Table S2).

5.2. Four-view loop detection on the four-storey sequence

The four-view pipeline retained loops on every floor. Of 1100 candidates, 207 had sufficient visual support, 42 agreed with the frontend odometry, and 29 passed the LiDAR checks. Sequence testing retained 28 pairs, from which 26 spatial anchors were selected. All 26 remained after graph optimization and were judged correct by visual inspection. Their distribution over F1–F4 was 11, 3, 7, and 5. Figure 4 shows their locations, including three in the occluded lateral stair area on F3.

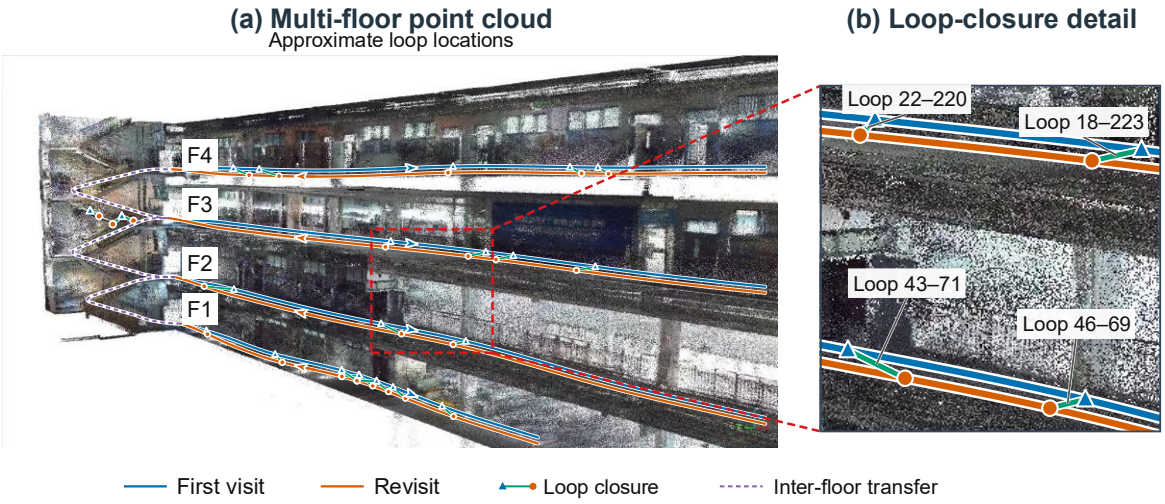


Figure 4. Four-storey point cloud with traversal routes and approximate locations of all 26 loops (F1–F4: 11/3/7/5). (a) First visits and revisits are shown in blue and orange; purple dashed lines indicate floor transfers. Three F3 loops are marked in the occluded lateral stair area. (b) Enlarged view of loops 22–220 and 18–223 on F3 and 43–71 and 46–69 on F2. Triangles and circles mark first-visit and revisit nodes, joined by green loop segments.

The forward pair produced the same number of candidates, and its shortlist included 14 of the 26 loops found by the four-view method. Five candidates had sufficient visual support, but all five differed from the frontend odometry by more than 5 m or 20°. No candidate reached the LiDAR stage (Table 1b). For this sequence, the loss of accepted loops occurred during visual pose verification.

Table 1. Component ablations on the single-storey and four-storey sequences.

(a) Single-storey LiDAR-verification ablation

Configuration	Correct	False	Total
Ours	17	0	17
w/o LiDAR verification	10	1	11

(b) Four-storey camera-input, ranking, and LiDAR-verification ablations

Stage or floor	Joint four-view	Forward pair	Independent max	Without LiDAR verification
Shortlisted candidates	1100	1100	—	1100
Sufficient visual support	207	5	—	—
Pose consistent with odometry	42	0	—	—
Pairs passed to sequence checking	29	0	—	42
Route-sequence supported	28	0	—	17
Spatial anchors	26	0	—	16
Accepted loops	26	0	20	16
Accepted on F1	11	0	8	10
Accepted on F2	3	0	5	0
Accepted on F3	7	0	7	6
Accepted on F4	5	0	0	0

(c) Single-storey sequence and GNC stress test

Configuration	Sequence	GNC	Spatial anchors	Accepted loops	Max position change (m)
Sequence + GNC	On	On	27	23	0.398
Sequence only	On	Off	27	27	46.441
GNC only	Off	On	50	26	1.029
Neither	Off	Off	50	50	95.173

Note. (a) Visual inspection classified accepted loops as correct or false. (b) Joint four-view and forward-pair configurations include bidirectional LiDAR verification. Without LiDAR verification passes the 42 visual and odometry-consistent pairs directly to sequence checking. Independent max changes candidate ranking. An em dash denotes a stage count not tabulated for that configuration. (c) The four configurations share 55 measurements and their covariances, with floor factors disabled. Maximum position change is the largest node displacement from the input trajectory. The main single-storey reconstruction contains 17 accepted loops.

Joint scoring also affected how loops were distributed along the route. Independent maximum-score ranking retained 20 loops, distributed as 8, 5, 7, and 0 over F1–F4 (Table 1b). Its shortlist omitted pair 191–168 on F4. The remaining F4 matches formed two groups of two pairs separated by 33.6 s. Each group fell below the three-pair sequence requirement, and the gap exceeded the 24 s limit. Joint ranking retrieved the

missing pair and retained five F4 loops. Fixed-camera ranking produced 12–15 loops, depending on the camera (Supplementary Table S1). The optimized maps from joint and independent ranking differed by at most 0.133 m in node position and 0.003 m in height; the main change was in loop coverage across floors.

The external configurations produced different loop sets (Table 2). BTC [17] proposed 172–226 candidates across the four tested parameter settings. None remained after the added same-floor, route-sequence, and odometry-consistency checks. The three tested RTAB-Map [11] configurations accepted 0, 11, and 10 loops. Four and three cross-floor associations were identified in the latter two configurations. The Scan Context [16] configuration accepted 35 loops, including two identified cross-floor pairs. Our pipeline accepted 26 visually confirmed loops distributed across all four floors.

Table 2. External loop-closure configurations on the four-storey sequence.

Method/configuration	Accepted loops	Cross-floor loops identified	Loops reviewed
BTC + proposed selection checks	0 after added checks	—	No accepted output
RTAB-Map, PnP	0	—	No accepted output
RTAB-Map, 3D–3D	11	4	Cross-floor cases
RTAB-Map, 3D–3D + 2.5 m	10	3	Cross-floor cases
Scan Context + ICP	35	2	Two cross-floor cases
Ours	26	0	All 26

Note. The BTC configuration combines BTC retrieval with our same-floor, route-sequence, and odometry-consistency checks. RTAB-Map and Scan Context use the configurations in Section 4.3. Visual review covered all 26 loops from our pipeline and the identified cross-floor cases from the external methods.

5.3. Staged verification of loop constraints

LiDAR verification affected both the accepted places and the pose measurements passed to later stages. On the single-storey sequence, removing this stage yielded 66 pairwise passes, 54 sequence-supported pairs, and 11 accepted loops. Visual inspection judged ten accepted loops correct and one false. The full pipeline retained 17 correct loops (Table 1a). Removing registration also changes the transformations evaluated by the sequence test and graph optimizer.

On the four-storey sequence, omitting the LiDAR stage increased the pairwise passes from 29 to 42. Sequence support then decreased from 28 to 17 pairs, and accepted loops decreased from 26 to 16 (Table 1b). The retained loops were distributed as 10, 0, 6, and 0 over F1–F4. Pair 125–91 was judged false and retained a GNC weight of 1.0. The full pipeline rejected this pair during LiDAR verification. GNC updates loop weights from graph residuals [20]. In this example, the wrong place association produced a measurement that remained compatible with those residuals.

The single-storey stress test examines sequence checking and GNC using the same permissive set of 55 measurements (Table 1c). Sequence support reduced the anchors from 50 to 27. With both sequence checking and GNC, 23 loops remained and the maximum node displacement was 0.398 m. Sequence checking alone retained all 27 anchors and produced a maximum displacement of 46.441 m. GNC alone retained 26 of 50 anchors, with a maximum displacement of 1.029 m. Disabling both checks increased the maximum displacement to 95.173 m. The two configurations without GNC produced large deformations relative to the approximately 0.2 m ground gap observed between corridor passes.

GNC removed four stress-test measurements: 73→54, 76→56, 77→57, and 78→58. Their endpoints were judged to depict the same places, but their relative poses conflicted with the remaining graph. This distinction explains why a correct place association can still produce a measurement that receives a low optimization weight. In the main four-storey experiment, enabling or disabling GNC retained the same 26 loops, with a maximum node-position difference of 0.042 mm.

Sequence-gap sensitivity is shown in Figure 5(a). Query-side limits of 12, 18, and 24 s produced 17, 21, and 28 supported pairs and 16, 19, and 26 accepted loops, respectively. At 12 s, neither F2 nor F4 received a loop. The counts remained at 28 supported pairs and 26 accepted loops for all tested limits from 25 to 48 s. We used 24 s for both recordings because it was the shortest evaluated setting on this plateau in the four-storey experiment.

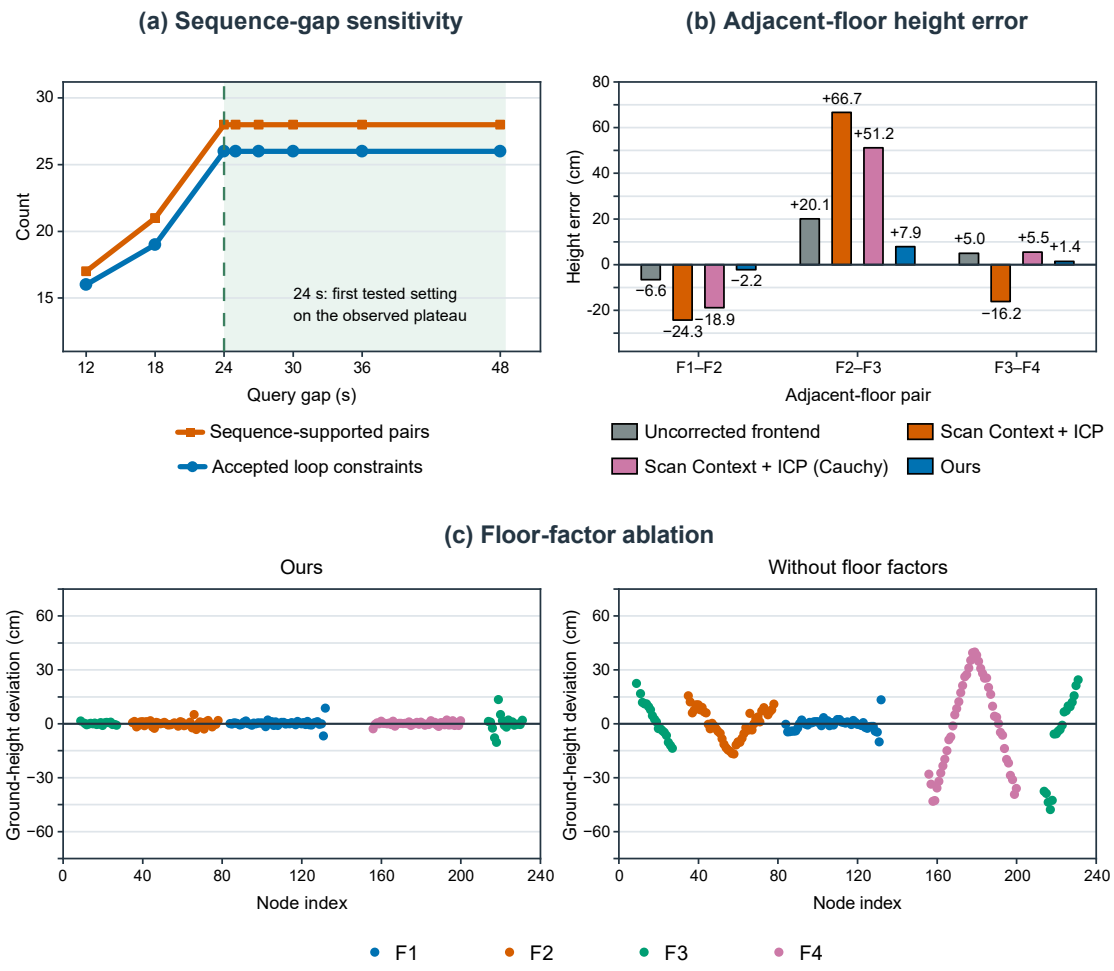


Figure 5. Sequence-gap sensitivity, adjacent-floor height error, and floor-factor ablation. (a) Query gaps of 12, 18, 24, 25, 27, 30, 36, and 48 s, with a historical-side limit of 1.5 times the query gap. At 24 s, 28 sequence-supported pairs yield 26 loop constraints. (b) Signed adjacent-floor height errors, computed as mapped ground separation minus the measured reference height. The reference heights for F1–F2, F2–F3, and F3–F4 are 4.201, 4.255, and 4.169 m, respectively. (c) Ground-height deviations for Ours and Without floor factors, using the same 26 loop constraints. Each ground-height observation is expressed relative to the median height of its physical floor within the corresponding reconstruction. Colours identify F1–F4.

5.4. Floor-aware optimization and vertical consistency

The floor constraints improved both adjacent-floor separation and ground-height consistency within each floor. The proposed method reconstructed F1–F2, F2–F3, and F3–F4 separations of 4.179, 4.334, and 4.183 m (Supplementary Table S4). Relative to the measured references, the signed errors were −2.2, +7.9, and +1.4 cm, giving a mean absolute error of 3.83 cm across the three pairs. With the same 26 loops and no floor factors, the separations were 4.075, 4.469, and 4.276 m, with a mean absolute error of 14.90 cm.

Within-floor P95–P5 ground-height spreads were 2.47, 4.29, 6.59, and 2.98 cm on F1–F4 with the floor factors. Removing them increased the spreads to 6.76, 26.18, 64.47, and 76.33 cm. Figure 5(c) shows the ground-height profiles on a common vertical scale. Because both reconstructions use the same loops, this comparison isolates the effect of the floor factors on vertical consistency.

The uncorrected frontend, Scan Context [16] with Gaussian loop noise, and Scan Context with Cauchy-robust loop noise had mean absolute separation errors of 10.57, 35.73, and 25.20 cm, respectively. The two Scan Context graphs contain the identified cross-floor loops and use their own loop measurements without floor factors. These results describe the complete reconstruction configurations; the comparison with the same 26 loops provides the direct floor-factor ablation.

5.5. Place recognition on a public dataset

The NCD Quad-Easy experiment examines four-camera retrieval on an outdoor handheld sequence, extending the evaluation beyond the indoor recordings [41, 42]. All methods use the same 31 queries. Ground-truth positive pairs satisfy a 2 m spatial threshold and a 30 s temporal exclusion. Table 3 reports both exact-node matches and matches with a tolerance of ±1 historical node. Exact matching requires the retrieved node itself to be a ground-truth positive. The tolerance rule also accepts its immediate neighbour in the historical sequence. With nodes approximately 3 s apart, that neighbour can lie beyond the 2 m spatial threshold.

Table 3. Place recognition on NCD Quad-Easy.

Retrieval method	**R@1, ±1 node**	**R@5, ±1 node**	**R@1, exact node**
Scan Context [16]	27/31 (87.1%)	28/31 (90.3%)	26/31 (83.9%)
DINOv2 mean-pool, four cameras [28]	29/31 (93.5%)	31/31 (100.0%)	16/31 (51.6%)
MixVPR, c0 [27]	28/31 (90.3%)	29/31 (93.5%)	19/31 (61.3%)
MixVPR max-sim, four cameras [27]	30/31 (96.8%)	31/31 (100.0%)	16/31 (51.6%)
Ours	30/31 (96.8%)	31/31 (100.0%)	17/31 (54.8%)

Note. All methods use the same 31 queries under the 2 m spatial threshold and 30 s temporal exclusion. The ±1-node rule accepts the reference node or an immediate historical neighbour; exact-node matching requires the retrieved node itself to be a ground-truth positive.

Under the ±1-node rule, our method retrieves a positive at rank one for 30 of 31 queries (96.8%) and within the top five for all 31 (100.0%). Four-camera MixVPR [27] achieves the same results. Scan Context [16], four-camera DINOv2 mean pooling [28], and single-camera MixVPR achieve R@1 values of 27/31, 29/31, and 28/31, respectively. The four-camera visual methods therefore retrieve an accepted match within the first five candidates for every query in this comparison.

Exact-node matching gives a different ordering. Scan Context achieves the highest R@1 at 26/31 (83.9%), followed by single-camera MixVPR at 19/31 (61.3%). Our method achieves 17/31 (54.8%), while four-camera DINOv2 mean pooling and four-camera MixVPR each achieve 16/31 (51.6%). For our method, the difference between the exact and tolerance-based scores shows that several first-ranked candidates are neighbouring nodes of the reference matches. Reporting both rules makes this difference in retrieval location visible.

After geometric verification and graph optimization, 100 loops remain. With a tolerance of ±1 keyframe at each endpoint, 96 of the 100 loops match ground-truth positive pairs, covering 32 of the 38 pairs. None of the five opposite-heading positive pairs is covered. The retrieval scores and the accepted-loop results describe successive stages of processing on this sequence. The Supplementary Note, "Public-dataset retrieval" gives the protocol and additional results.

6. Discussion and Limitations

6.1. Discussion

The indoor results show how surrounding camera observations contribute to loop closure during a return traversal. A change in heading can move the useful view of a surface from one camera to another. The forward-pair shortlist already contained several of the loops retained by the full method, but those candidates failed during pose verification. Cyclic camera assignment connects the relevant views, and the common-frame pose estimate combines their correspondences. Passing the selected assignment to verification lets both stages use the corresponding views from the two visits.

The ranking ablation also shows why the arrangement of candidate pairs along the route matters. On F4, one additional retrieved pair joined observations that otherwise formed short, disconnected groups. That change supplied the sequence support needed to retain loops on the floor. The final maps from joint and independent ranking remained close, while their accepted loops covered different parts of the building. Loop distribution therefore adds useful information to the total loop count when assessing the retrieval stage.

The verification stages evaluate different aspects of a measurement. LiDAR checking tests the local submap alignment, sequence checking compares corrections along the route, and GNC evaluates the loop within the full graph. Their roles are visible in the ablations: a false place association survived graph weighting when LiDAR verification was removed, while some correctly associated stress-test pairs were rejected because their relative poses disagreed with the graph. In the main four-storey run, the measurements reaching optimization were sufficiently consistent that GNC changed the solution only slightly.

Ground observations extend the correction across the floor surfaces. Each observation is attached to an individual pose and to the shared height of its floor, so both revisited and singly traversed segments contribute to the vertical estimate. The fixed-loop ablation shows the effect of this shared reference through the reduction in within-floor height spread. Connections between floor levels retain the measured differences in storey separation. Together, these constraints link the corrected trajectory to the vertical structure visible in the ground observations.

6.2. Limitations

The two indoor recordings were acquired in one building, and their data were used to set the thresholds, including the sequence-gap limit. Evaluation in further buildings and along different traversal routes is needed to assess how these settings transfer. Floor identification also depends on the initial trajectory and

usable ground-plane fits. The missed opposite-heading pairs on NCD motivate additional tests of reverse traversal under different scene and acquisition conditions.

Accepted place associations were reviewed by one author. Independent loop labels and reference relative poses would allow evaluation of overall loop recall and relative-pose accuracy. The reported surface gaps and ground-height spreads describe map consistency, while the measured storey separations provide an external vertical reference. External-method comparisons use their stated pipeline configurations, including different verification procedures and floor priors.

The current backend processes a completed recording offline. Runtime on the acquisition platform has not been measured. An incremental implementation would require evaluating computation time alongside the arrival of image, LiDAR, and ground observations.

7. Conclusions

This study presented a visual–LiDAR loop-closure backend for multi-storey indoor mapping across repeated corridor passes. Four-camera retrieval establishes correspondences between viewing directions at different visits, and image matches with LiDAR depth provide relative-pose measurements. The accepted measurements and ground observations are then combined in a floor-aware pose graph to correct the reconstructed map.

Map correction was evaluated at corridor and building scales. On the single-storey sequence, the median ground gap between repeated corridor observations decreased from 21.00 to 0.39 cm, accompanied by smaller upper-surface and wall-normal gaps. On the four-storey sequence, the method retained 26 visually confirmed loops across every floor, while the forward pair retained none. Using the same loop set, floor constraints reduced ground-height variation on all four floors and lowered the mean absolute adjacent-floor separation error from 14.90 to 3.83 cm.

These results connect successful revisit detection with measurable changes in the final reconstruction. Cross-camera observations supplied loop measurements from the indoor return passes, and shared floor heights improved the consistency of the building's vertical structure. Across the two recordings, the combined correction brought repeated surfaces and floor levels into closer agreement while retaining the original point observations. Surrounding visual coverage connects reverse revisits, and shared floor geometry constrains the vertical structure during map correction.

Data availability

Code, configurations, experimental results, trajectories, loop-closure records, and figure and table data are publicly available at https://github.com/yuanbanlin8-dotcom/four-view-visual-lidar-loop-closure (accessed on 30 October 2026). The Supplementary Methods section "Reproducibility" provides the reproduction instructions. The Newer College source data are available through the dataset authors.

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Additional information

Competing interests

The authors declare no competing interests.